

STEP 1 — Install Required Libraries

```
!pip install gensim gdown scikit-learn matplotlib
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: gdown in /usr/local/lib/python3.12/dist-packages (5.2.1)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.0)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (from gdown) (4.13.5)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from gdown) (3.20.3)
Requirement already satisfied: requests[socks] in /usr/local/lib/python3.12/dist-packages (from gdown) (2.32.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from gdown) (4.67.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.0)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->gdown) (2.8.3)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->gdown) (4.12.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (2025.1.1)
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown) (1.7.1)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
27.9/27.9 MB 26.0 MB/s eta 0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
!gdown --id 0B7XkCwpISKDYN1NUTT1SS21pQmM
```

```
/usr/local/lib/python3.12/dist-packages/gdown/__main__.py:139: FutureWarning: Option '--id' was deprecated in version 4.3.1
  warnings.warn(
Downloading...
From (original): https://drive.google.com/uc?id=0B7XkCwpISKDYN1NUTT1SS21pQmM
From (redirected): https://drive.google.com/uc?id=0B7XkCwpISKDYN1NUTT1SS21pQmM&confirm=t&uuiid=1dbf8e0e-05da-4923-850a-6acd8
To: /content/GoogleNews-vectors-negative300.bin.gz
100% 1.65G/1.65G [00:21<00:00, 76.0MB/s]
```

```
!gzip -d GoogleNews-vectors-negative300.bin.gz
```

Import Required Libraries

```
import numpy as np # Numerical operations on vectors
from gensim.models import KeyedVectors # Load pre-trained embeddings
from sklearn.metrics.pairwise import cosine_similarity # Similarity calculation
from sklearn.decomposition import PCA # Dimensionality reduction
import matplotlib.pyplot as plt # Visualization
```

STEP 2: Load Pre-trained Word Embeddings

```
model = KeyedVectors.load_word2vec_format(
    "GoogleNews-vectors-negative300.bin",
    binary=True,
    limit=200000 # limits vocab to speed up loading (optional)
)

print("Vocabulary size:", len(model))
print("Vector dimension:", model.vector_size)
```

```
Vocabulary size: 200000
Vector dimension: 300
```

```
word = "king"
vector = model[word]

print(f"Word vector for '{word}':")
print(vector)
print("Vector length:", len(vector))
```

```
Word vector for 'king':
[ 1.25976562e-01  2.97851562e-02  8.60595703e-03  1.39648438e-01
 -2.56347656e-02 -3.61328125e-02  1.11816406e-01 -1.98242188e-01
 5.12695312e-02  3.63281250e-01 -2.42187500e-01 -3.02734375e-01
 -1.77734375e-01 -2.49023438e-02 -1.67968750e-01 -1.69921875e-01
 3.46679688e-02  5.21850586e-03  4.63867188e-02  1.28906250e-01
 1.36718750e-01  1.12792969e-01  5.95703125e-02  1.36718750e-01
 1.01074219e-01 -1.76757812e-01 -2.51953125e-01  5.98144531e-02
 3.41796875e-01 -3.11279297e-02  1.04492188e-01  6.17675781e-02
 1.24511719e-01  4.00390625e-01 -3.22265625e-01  8.39843750e-02
 3.90625000e-02  5.85937500e-03  7.03125000e-02  1.72851562e-01
 1.38671875e-01 -2.31445312e-01  2.83203125e-01  1.42578125e-01
 3.41796875e-01 -2.39257812e-02 -1.09863281e-01  3.32031250e-02
 -5.46875000e-02  1.53198242e-02 -1.62109375e-01  1.58203125e-01
 -2.59765625e-01  2.01416016e-02 -1.63085938e-01  1.35803223e-03
 -1.44531250e-01 -5.68847656e-02  4.29687500e-02 -2.46582031e-02
 1.85546875e-01  4.47265625e-01  9.58251953e-03  1.31835938e-01
 9.86328125e-02 -1.85546875e-01 -1.00097656e-01 -1.33789062e-01
 -1.25000000e-01  2.83203125e-01  1.23046875e-01  5.32226562e-02
 -1.77734375e-01  8.59375000e-02 -2.18505859e-02  2.05078125e-02
 -1.39648438e-01  2.51464844e-02  1.38671875e-01 -1.05468750e-01
 1.38671875e-01  8.88671875e-02 -7.51953125e-02 -2.13623047e-02
 1.72851562e-01  4.63867188e-02 -2.65625000e-01  8.91113281e-03
 1.49414062e-01  3.78417969e-02  2.38281250e-01 -1.24511719e-01
 -2.17773438e-01 -1.81640625e-01  2.97851562e-02  5.71289062e-02
 -2.89306641e-02  1.24511719e-02  9.66796875e-02 -2.31445312e-01
 5.81054688e-02  6.68945312e-02  7.08007812e-02 -3.08593750e-01
 -2.14843750e-01  1.45507812e-01 -4.27734375e-01 -9.39941406e-03
 1.54296875e-01 -7.66601562e-02  2.89062500e-01  2.77343750e-01
 -4.86373901e-04 -1.36718750e-01  3.24218750e-01 -2.46093750e-01
 -3.03649902e-03 -2.11914062e-01  1.25000000e-01  2.69531250e-01
 2.04101562e-01  8.25195312e-02 -2.01171875e-01 -1.60156250e-01
 -3.78417969e-02 -1.20117188e-01  1.15234375e-01 -4.10156250e-02
 -3.95507812e-02 -8.98437500e-02  6.34765625e-03  2.03125000e-01
 1.86523438e-01  2.73437500e-01  6.29882812e-02  1.41601562e-01
 -9.81445312e-02  1.38671875e-01  1.82617188e-01  1.73828125e-01
 1.73828125e-01 -2.37304688e-01  1.78710938e-01  6.34765625e-02
 2.36328125e-01 -2.08984375e-01  8.74023438e-02 -1.66015625e-01
 -7.91015625e-02  2.43164062e-01 -8.88671875e-02  1.26953125e-01
 -2.16796875e-01 -1.73828125e-01 -3.59375000e-01 -8.25195312e-02
 -6.49414062e-02  5.07812500e-02  1.35742188e-01 -7.47070312e-02
 -1.64062500e-01  1.15356445e-02  4.45312500e-01 -2.15820312e-01
 -1.11328125e-01 -1.92382812e-01  1.70898438e-01 -1.25000000e-01
 2.65502930e-03  1.92382812e-01 -1.74804688e-01  1.39648438e-01
 2.92968750e-01  1.13281250e-01  5.95703125e-02 -6.39648438e-02
 9.96093750e-02 -2.72216797e-02  1.96533203e-02  4.27246094e-02
 -2.46093750e-01  6.39648438e-02 -2.25585938e-01 -1.68945312e-01
 2.89916992e-03  8.20312500e-02  3.41796875e-01  4.32128906e-02
 1.32812500e-01  1.42578125e-01  7.61718750e-02  5.98144531e-02
 -1.19140625e-01  2.74658203e-03 -6.29882812e-02 -2.72216797e-02
 -4.82177734e-03 -8.20312500e-02 -2.49023438e-02 -4.00390625e-01
 -1.06933594e-01  4.24804688e-02  7.76367188e-02 -1.16699219e-01
 7.37304688e-02 -9.22851562e-02  1.07910156e-01  1.58203125e-01
 4.24804688e-02  1.26953125e-01  3.61328125e-02  2.67578125e-01
 -1.01074219e-01 -3.02734375e-01 -5.76171875e-02  5.05371094e-02
 5.26428223e-04 -2.07031250e-01 -1.38671875e-01 -8.97216797e-03
 -2.78320312e-02 -1.41601562e-01  2.07031250e-01 -1.58203125e-01
 1.27929688e-01  1.49414062e-01 -2.24609375e-02 -8.44726562e-02
```

STEP 3: Explore Word Similarity

```
word_pairs = [
    ("doctor", "nurse"),
    ("cat", "dog"),
    ("car", "bus"),
    ("king", "queen"),
    ("apple", "banana"),
    ("computer", "keyboard"),
    ("teacher", "student"),
    ("city", "village"),
    ("sun", "moon"),
    ("book", "library")
]

for w1, w2 in word_pairs:
    similarity = model.similarity(w1, w2)
```

```
print(f"Similarity({w1}, {w2}) = {similarity:.4f}")
```

```
Similarity(doctor, nurse) = 0.6320
Similarity(cat, dog) = 0.7609
Similarity(car, bus) = 0.4693
Similarity(king, queen) = 0.6511
Similarity(apple, banana) = 0.5318
Similarity(computer, keyboard) = 0.3964
Similarity(teacher, student) = 0.6301
Similarity(city, village) = 0.4790
Similarity(sun, moon) = 0.4263
Similarity(book, library) = 0.3245
```

STEP 4:Nearest Neighbor Exploration

```
query_words = ["king", "university", "computer", "hospital", "music"]

for word in query_words:
    print(f"\nTop similar words to '{word}':")
    for similar_word, score in model.most_similar(word, topn=5):
        print(f"{similar_word} ({score:.4f})")
```

Top similar words to 'king':

```
kings (0.7138)
queen (0.6511)
monarch (0.6413)
crown_prince (0.6204)
prince (0.6160)
```

Top similar words to 'university':

```
universities (0.7004)
faculty (0.6781)
undergraduate (0.6587)
campus (0.6435)
college (0.6385)
```

Top similar words to 'computer':

```
computers (0.7979)
laptop (0.6640)
laptop_computer (0.6549)
Computer (0.6473)
laptop_computers (0.5585)
```

Top similar words to 'hospital':

```
Hospital (0.7932)
hospitals (0.7213)
intensive_care (0.7206)
Intensive_Care_Unit (0.6634)
Medical_Center (0.6558)
```

Top similar words to 'music':

```
classical_music (0.7198)
jazz (0.6835)
Music (0.6596)
songs (0.6396)
musicians (0.6336)
```

STEP 5:Word Analogy Tasks

```
analogies = [
    ("king", "man", "woman"),
    ("paris", "France", "india"),
    ("teacher", "school", "hospital")
]

for a, b, c in analogies:
    result = model.most_similar(
        positive=[a, c],
        negative=[b],
        topn=1
    )
    print(f"{a} - {b} + {c} = {result[0][0]}")
```

```
king - man + woman = queen
paris - France + india = indian
teacher - school + hospital = Hospital
```

STEP 6: Visualization

```
words = [
    "king", "queen", "man", "woman",
    "doctor", "nurse", "hospital",
    "teacher", "student", "school",
    "car", "bus", "train",
    "apple", "banana", "orange"
]

vectors = np.array([model[word] for word in words])

pca = PCA(n_components=2)
reduced_vectors = pca.fit_transform(vectors)

plt.figure(figsize=(10, 8))
plt.scatter(reduced_vectors[:, 0], reduced_vectors[:, 1])

for i, word in enumerate(words):
    plt.annotate(word, (reduced_vectors[i, 0], reduced_vectors[i, 1]))

plt.title("Word Embedding Visualization using PCA")
plt.show()
```



